



His Hands Can't Hit What His Eyes Can't See

Emergent AI Tactics for Successful Operations in the Air Littoral

Heiko Borchert and Christian Brandlhuber

DAIO Study 24|26

Ein Projekt im Rahmen von

dtec.bw
Zentrum für Digitalisierungs- und
Technologieforschung der Bundeswehr



Finanziert von der
Europäischen Union
(Euklidischer Raum)

About the Defense AI Observatory

The Defense AI Observatory (DAIO) at the Helmut Schmidt University in Hamburg monitors and analyzes the use of artificial intelligence by armed forces. DAIO comprises three interrelated work streams:

- Culture, concept development, and organizational transformation in the context of military innovation
- Current and future conflict pictures, conflict dynamics, and operational experience, especially related to the use of emerging technologies
- Defense industrial dynamics with a particular focus on the impact of emerging technologies on the nature and character of techno-industrial ecosystems

DAIO is an integral element of GhostPlay, a capability and technology development project for concept-driven and AI-enhanced defense decision-making in support of fast-paced defense operations. GhostPlay is funded by the Center for Digital and Technology Research of the German Bundeswehr (dtec.bw). dtec.bw is funded by the European Union – NextGenerationEU.

Ein Projekt im Rahmen von



His Hands Can't Hit What His Eyes Can't See

Emergent AI Tactics for Successful Operations in the Air Littoral

Heiko Borchert and Christian Brandlhuber

DAIO Study 24|26

Ein Projekt im Rahmen von

 **dtec.bw**
Zentrum für Digitalisierungs- und
Technologieforschung der Bundeswehr



About the Authors

Dr. Heiko Borchert is DAIO's co-director and runs a strategic affairs consulting boutique. His work portfolio spans a broad spectrum of topics including defense exports, defense research, multinational armaments cooperation and maritime security. Dr. Borchert studied international relations, business administration, economics, and law at the University of St. Gallen, where he also got his Ph.D.

Dr. Christian Brandlhuber is co-founder and CTO of 21strategies. He studied mathematics and computer science at the TU Munich and the TH Karlsruhe. He has been involved in developing AI solutions for defense, industry, and financial market applications since the late 1990s. Dr. Brandlhuber is co-holder of five granted patents, inter alia, in the fields of data fusion and holds a granted patent for advanced financial planning systems.

Acknowledgments

We gratefully acknowledge the contribution of Lennart Heggmann, Raphael Jakoby, Dr. Duarte Magano and Dr. Scott Muller as well as the Germany Army Concepts and Capability Development Center in writing this paper. None of the statements in this report should be interpreted as to construe official positions of the Center.

Design

Almasy Information Design Thinking

Imprint

Heiko Borchert and Christian Brandlhuber, His Hands Can't Hit What His Eyes Can't See. Emergent AI Tactics for Successful Operations in the Air Littoral DAIO Study 24/26 (Hamburg: Defense AI Observatory, 2024).

Defense AI Observatory | Chair of Political Theory | Helmut Schmidt University
Holstenhofweg 85 | 22043 Hamburg | T +49 40 6541 2776
www.defenseai.eu | contact@defenseai.eu | @Defense_AIO

ISSN (online): 2749-5337

ISSN (print): 2749-5345

Content

1 Summary	6
2 Boxing Legends on the Battlefield	8
3 Defense Metaverse: A New Digital Environment for Force Development	11
4 Wild Hornets: Fine Tuning the Defense Metaverse for Advanced Concepts Development.....	16
4.1 Fighting in the Air Littoral: Setting the Scene	17
4.2 Wild Hornets Parameters.....	18
5 Emergent AI Tactics Make the Difference	23
5.1 Setting the Baseline.....	24
5.2 AI Tactics for the Endgame	25
5.3 AI Tactics for Reconnaissance-Strike Missions	31
5.4 AI Tactics for Target Assignment	33
6 Outlook: An AI Tactics Kernel for Software-Defined Defense	35
Literature.....	38

1 Summary

Mass alone is insufficient to overwhelm adversaries. This is the core message of our paper. But wait. Before critics of the 'quantity is a quality of its own' narrative prematurely rejoice, both nuance and context must be grasped to fully understand what we mean.

One of the key problems with the 'mass matters' movement is that the adversary has a say on the battlefield – and how adversaries respond to attacks with a high number of unmanned systems, for example, very much determines the outcome of the encounter. But response is not a one-way street. How attackers react to a first round of countermeasures very much determines if and to what extent they can accomplish their mission. So, a different approach to concepts and capability development is needed to gauge if, how, and when mass is decisive and how the offense vs defense game is likely to play out.

Developing and implementing this new approach is at the heart of Wild Hornets, an advanced concepts study conducted in cooperation with the German Army Concepts and Capability Development Center (ACCDC). Over the past 18 months we have analyzed how Army aviators flying attack and transport helicopters could best use so-called Air-Launched Effects (ALE) to identify and neutralize adversarial Ground-Based Air Defense (GBAD) constellations. To this purpose we have developed a high-fidelity Defense Metaverse, a virtual twin of the battlefield. The Defense Metaverse replicates all parameters of the mission-relevant area of interest to digitally model and simulate the interaction between ALE and GBAD systems as well as the relevant sensors and effectors.

Our Defense Metaverse uses counter-play logic and dynamics to let red and blue forces interact. This, we argue, is the only way for military concept developers to get a sufficient understanding of the tactical added value that different building blocks of future military systems might provide. Furthermore, this approach illustrates that, in interaction with intelligent adversaries, artificial intelligence (AI) can self-learn complex tactical behavior. This pushes the concepts and technology envelope beyond current patterns of refining existing technologies with AI. AI tactics open the door to novel behavior on the battlefield to surprise the opponent and gain the upper hand.

Thus, it is not mass, but smart mass that matters – and AI tactics shape smartness. As we demonstrate, self-learned collaborative coordination among the members of a reconnaissance-strike swarm of unmanned systems leverages the principles of war to provide a major leap forward. This approach goes well beyond current swarm concepts. Traditionally, swarm members execute pre-planned flight trajectories and

use relatively simple rules to keep distance from each other, maintain formation, or implement rudimentary cooperation patterns. By contrast, our approach uses distributed control to facilitate cooperation among swarm members to overwhelm adversarial air defense postures. This means that each swarm member follows its own goals and learns how to cooperate with teammates. As a result, cooperation is a function of individual benefit analyses: Swarm members will cooperate, if giving up individual objectives will help them accomplish a common goal of higher value.

These findings propel military performance to new levels as AI tactics that enable self-learned collaborative coordination

- *enhance efficiency* because fewer ALE are needed to engage a GBAD constellation, meaning that – potentially – more targets can be attacked with a given number of unmanned assets;
- *improve effectiveness* by enabling ALE swarms to adjust mission tempo to mission requirements and thus opt for low and slow flight paths to sneak into the GBAD's 'dead zone' for effective engagement;
- *unlocks emergence* as the ability to instantly respond to unforeseen adversarial moves by defining and delegating tasks and subtasks among members of the swarm at campaign speed.

These outcomes are paramount for missions in the air littoral that constitute the core interest of Wild Hornets. This small air strip is increasingly contested. Control of the air littoral is essential to deliver air power that too often only focuses on blue skies operations while at the same time ensuring 'protection from above' for land forces. The AI tactics discussed in this paper illustrate how even very small groups of swarms with only four or five team members can effectively engage even powerful gun-based short-range GBAD solutions like the Gepard system. At the same time, AI tactics also demonstrate how these defender systems can be upgraded to withstand AI-empowered swarms.

Based on these findings we argue that an AI tactics kernel will constitute a key building block for future software-defined defense solutions. This tactics kernel can be used as a multi-avenue transmission mechanism to combine current capability demands with ongoing battlefield requirements and visions of future forces.

This is the second paper in a trilogy of white papers addressing the role of AI tactics on the battlefield. These papers grow out of GhostPlay, a capability and technology development project aimed at developing red and blue AI tactics. *Free Jazz on the Battlefield*, our first paper published in 2022, laid out the core idea, described key aspects of the basic method – for example regarding the use of decentralized and non-hierarchical command and control – and presented our initial findings focusing on the role of air defense against aggressor swarms. Later this year, we will release our third paper addressing AI tactics for naval combat.

2 Boxing Legends on the Battlefield

*Float like a butterfly, sting like a bee;
his hands can't hit what his eyes can't see*
Muhammed Ali¹

Everybody has a plan 'till they got punched in the mouth
Mike Tyson²

In 2014, then Deputy U.S. Secretary of Defense Robert O. Work and Shawn Brimley were among the first to argue that swarms of unmanned systems are a potential game-changer for future warfighting. Mass, they argued, would provide a new option to overwhelm adversaries and gain the upper hand.³ Since then, 'quantity as quality' has increasingly been interpreted as a leitmotiv for force development in NATO/EU countries and beyond. Most recently, Deputy U.S. Secretary of Defense Kathleen Hicks presented the Replicator initiative that leverages mass for future warfighting.⁴ Along this line, the United States have approved the sale of 291 Altius 600M-V loitering munitions and 720 Switchblade 300 to backup Taiwan's defense posture.⁵ Russia's use of Shahed-136 drones and other unmanned systems in Ukraine⁶ as well as the April 2024 deployment of 300 unmanned assets and missiles by Iran against Israel⁷ seem to vindicate the 'quantity as quality' narrative.

Unmanned mass can play an important role to fight a capable peer competitor, for example, by distributing force elements across domains and theaters of operations. But – and this is the key question for force developers – is it as decisive as presumed?

Muhammed Ali and Mike Tyson, the two great boxing champions quoted above, lend a proverbial 'helping hand' in answering this question. First, the so-called kill chain⁸ that combines sensors and shooters to identify and hit targets has come to play a foundational role in military thinking. But Ali perfectly captured the sensor-to-shooter dilemma when pointing out that the performance of the effectors very much depends on available sensors – and the latency of data and information processing that connects the two. Therefore hands (effectors) can only hit, what eyes (sensors) can see. Mass alone does not provide an easy option out of this dilemma. Rather, mass increases the challenge to smoothly integrate and coordinate all elements of this sensor-to-shooter cycle.

1 Johnston, "Muhammad Ali's best quotes."

2 Freedman, *Strategy*, p. ix.

3 Work/Brimley, 20YY.

4 Hicks, "'Unpacking the Replicator Initiative' at the Defense News Conference."

5 Arthur, "US approves loitering munitions sale for Taiwan's 'porcupine strategy.'"

6 Pettyjohn, *Evolution Not Revolution*.

7 Magid/Spiro, "April 13: Iran fires 300 missiles and drones; most intercepted; minor damage at IDF base."

8 Brose, *The Kill Chain*.

Second, mass does provide new options to deliver military power, but mass also adds extra burden on force and logistics planners. Mosaic warfare,⁹ that foresees a deconstruction of force into single units and a reconfiguration of these units at campaign speed and commensurate with mission requirements, provides the basis for mass to become a decisive factor, but traditional planning is not fit for the 'Tyson test.' Planning can only go so far as the knowledge about red and blue forces available at the time of initiating the planning cycle. Spontaneous punches by the enemy, as Tyson rightly observed, cannot be foreseen. Again, only increasing the number of (unmanned) assets without changing the paradigm to coordinate these assets will likely render force engagement against adversaries more cumbersome. Therefore, we argue that there is a pressing need to shift to approaches that are much more cognizant of how to respond to unforeseeable events.

Consequently, caution is at place. Together with the Germany Army Concepts and Capability Development Center (ACCDC) we have been working on Wild Hornets, an 18-month advanced concepts study. We have built a tailored Defense Metaverse, a high-fidelity virtual twin of the battlefield that meets the requirements of the Center, to develop AI tactics for future concepts on the helicopter use of Air-Launched Effects (ALE), such as Switchblade 600, against Ground-Based Air Defense (GBAD). Based on the findings summarized in this paper, we conclude: On its own, unmanned mass is not decisive. Rather, for mass to prevail, emergent AI tactics, emphasizing formation, not formulation,¹⁰ are needed – in particular when meeting an AI-augmented GBAD equipped with powerful countermeasures to fend off attacking ALE swarms.

Our cautionary note on unmanned mass as the new go-to solution for armed forces is important. It highlights that using the Defense Metaverse for concept development can prevent military planners from being misled by popular ideas. In addition, our study also illustrates the opportunities that software-defined defense offers to come up with new concepts, specifications for future defense products as well as novel in-service and military training ideas to prepare armed forces for a more assertive, more volatile, and thus also more risk-prone strategic environment.

9 Grayson, "Mosaic Warfare," Haystead, "DARPA's Mosaic Warfare; Clark/Patt/Schramm, Mosaic Warfare.

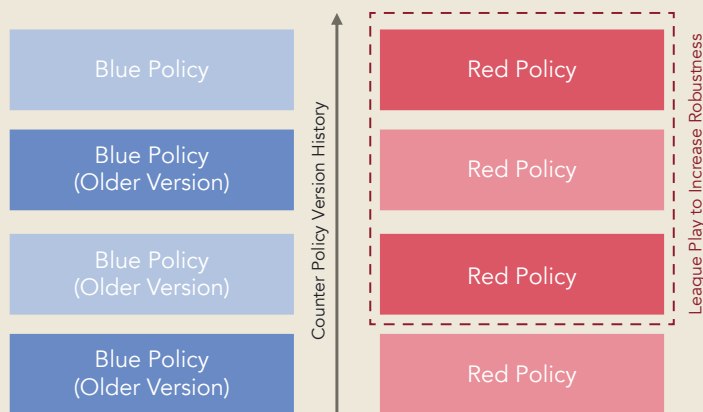
10 Borchert/Brandhuber/Brandstetter/Schaal, *Free Jazz on the Battlefield*, pp. 12-17; Borchert, "The Very Long Game of Defense AI Adoption: Introduction," p. 7.

3 Defense Metaverse: A New Digital Environment for Force Development

The Defense Metaverse is a high-fidelity virtual twin of the battlefield. Neal Stephenson first coined the term metaverse in 1992 to describe a three-dimensional virtual space in which humans interact with software agents.¹¹ Since then the concept has come to describe the idea of creating a digital replica for physical objects and processes to interact. Civil industry has embraced the concept as Industry 4.0 to denote the comprehensive use of digitalization to bring all stages of the product life cycle – from research and development to maintenance and sustainment – into a virtual environment. Defense companies are catching up. What started with individual digital models used for specific simulations, for example to assess the consequences of explosives against different types of armor, has become more common practice with the help of digital model-based engineering.¹²

The Defense Metaverse, that has been established with GhostPlay, follows these lines of effort but offers unique features and capabilities. Our Defense Metaverse creates a highly realistic replica of the battlefield, illustrating, for example, sensor and effector models, weather conditions, vegetation, terrain as well as urban and rural infrastructure. Most importantly, we use the counterplay logic (Figure 1) to have artificial intelligence (AI) develop tactics – not tactics modelled on human assumptions, but tactics completely self-learned by AI. Thus, we let AI play ‘free jazz’ to explore what tactics are best suited to achieve mission success within given specific parameters set by humans.

Figure 1: GhostPlay's Counterplay Logic



¹¹ Stephenson, Snow Crash.

¹² Digital Engineering Strategy, p. 3.

Counterplay enables AI to develop successful tactics in interaction with a smart adversary. Technically speaking, the system is forward planning a series of play moves to determine the optimal decision sequence. As soon as one actor has developed successful policies, the opponent learns to respond. Counterplay's goal is to gradually improve the performance of action and decision rules by training them against more complex and more challenging adversaries. In the end, this approach helps define non-exploitable policies.

Counterplay within the Defense Metaverse creates a new opportunity to develop scenarios almost without scripting, as interaction replaces scripting. This provides two key advantages:

- First, it is possible to cover a significantly larger number of scenarios than traditional simulations can handle. This includes scenarios illustrating unexpected and surprising behavior, which are most beneficial for concepts development, to assess technical specifications and set the parameters for worst, best, and expected case analyses. While the Defense Metaverse offers AI to operate in the 'free jazz' mode without any restrictions, it is also possible to set limits for red forces only to engage within specific and known rules of engagement or to mimic known tactical behavior.
- Second, counterplay significantly reduces the risk of analytical bias, i.e., design choices for the simulation that are based on specific beliefs/assumptions about adversarial action. Analyst bias is problematic as it may entail conceptual traps, that will normally not be questioned as the respective hypotheses are considered well-established knowledge. But reconfirming what is already known prevents the emergence of non-conventional behavior that would challenge existing wisdom. Red force tactics developed with counterplay, by contrast, produce unexpected outcomes, which may be most important to identify design flaws in workflows of systems used by defenders.

GhostPlay's approach to counterplay also underpins the development of decentralized control strategies, a major element of Mosaic Warfare.¹³ As we will discuss later, decentralized control strategies imply that, for example, members of a swarm of unmanned systems learn to develop collaborative tactics to accomplish a mission in tandem. This, in turn, enables the Defense Metaverse to run high-intensity combat simulations including a large number of objects that can change their behavior commensurate with the evolution of the simulated scenario – thus pushing the envelope for situations in which blue or red force behavior changes the game in one or the other direction.

13 See also: Clark/Patt/Walton, Implementing Decision-Centric Warfare.

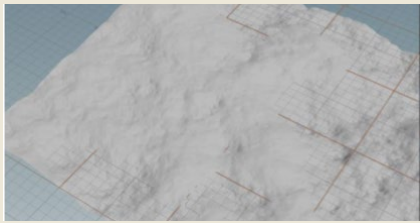
It goes without saying that for the Defense Metaverse to offer these simulation options, the digital environment needs to be modeled very realistically. As we discuss the specific parameters relevant for the Wild Hornets concept study in the following section, we want to spend a moment on describing the efforts needed to model the operational environment.

Tactical assertiveness, as outlined in this paper, requires the simulation to process environmental data like terrain, buildings, vegetation, weather, and views. To develop the required level of detail in a relatively short span of time, we have created automated pipelines. These pipelines use different data to generate a digital model of the target area under consideration (Figure 2).

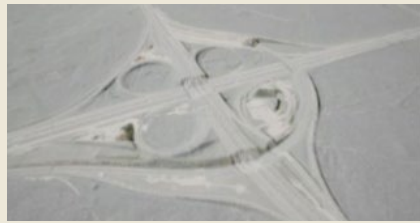
Building the pipelines starts with using Digital Terrain Elevation Data (DTED), which is georeferenced. The missions analyzed for Wild Hornets were set to play in Franconia, Bavaria (Germany), a region that is quite typical for Central Europe. Information on buildings augmented DTED, and both were fused with Light Detection and Ranging (LIDAR) data. This helped create a terrain and building data profile with a granularity of up to 10cm. We have synthetically generated vegetation to realistically replicate the type and cover of vegetation in the mission area. Depending on the weather conditions to be simulated, trafficability analyses can be conducted.

In preparing data for Wild Hornets, we benefitted from the fact that Bavarian authorities offer open-source digital terrain data. This, however, won't be the case for many other areas. That's why we have developed a specific development pipeline based on photogrammetric procedures and Neural Radiance Fields (NeRF) to reconstruct and model three-dimensional infrastructure data based on remote sensing.

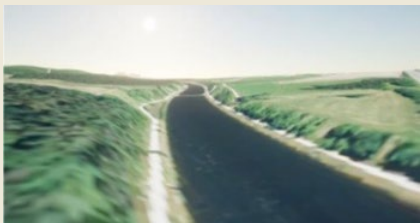
Figure 2: Modelling the Operational Environment



Digital Terrain Elevation Data (DTED) from DGM1 data (1x1 km tiles) are combined to deliver resolution of 1 m over the area of interest



Terrain data come with artefacts, which need to be cleaned



Texture can be obtained by digital orthophotos DOP in 0.4m resolution



Structure of buildings is reconstructed based on public registry data, including type of building material and other features as available



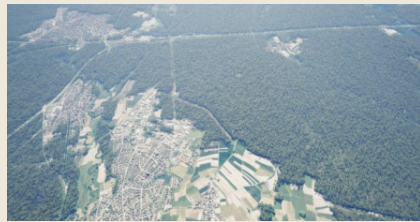
Fusion with LiDAR data scans increases resolution to 0.1m



Building material is used to train sensors. Roads are analyzed to gauge their weight capacity; weather-dependent trafficability is estimated



Use of "computational botanics" creates vegetation plant by plant, generating an accurate digital twin. As a result, realistic forest pass-through is possible while sensors scan realistic ground clutter



Weather and light effects create a realistic environment, which enables learning, for example, with mappings based on real sensor readings

4 Wild Hornets: Fine Tuning the Defense Metaverse for Advanced Concepts Development

Wild Hornets is an advanced concepts study conducted in cooperation with the German Army Concepts and Capability Development Center. The purpose of the study is to evaluate future capability requirements for so-called Air-Launched Effects (ALE), unmanned assets occupying the air space between traditional Unmanned Aerial Systems (UAS) and missiles. In doing so, the study went beyond off-the-shelf solutions to assess the tactical added value these systems offer in operations against highly performant Ground-Based Air Defense Systems (GBAD). One goal of the analysis was to scrutinize if and to what extent specific technical features such as the ability to exploit terrain features for low and lowest flight paths add tactical value. The second goal was to develop new tactics, techniques, and procedures (TTP) to use ALE. While evaluating existing TTP defined by the Army Center's experts, we also used AI to develop new TTPs for aggressor swarms and defender GBAD systems.

Before addressing the specific technical parameters set for Wild Hornets, we start by describing the challenges of fighting in the air littoral. This domain is pivotal to ensure the freedom of maneuver of allied forces and deny the enemy the flexibility to change battlefield positions quickly and safely. Lethality, detectability, survivability, and tempo all play a key role in the air littoral – and AI is instrumental in shaping the interplay between these key parameters.

4.1 Fighting in the Air Littoral: Setting the Scene

The air littoral is an increasingly important warfighting domain, but what winning in this narrow air corridor is all about has been under-conceptualized. The air littoral constitutes the 'air bridge' between blue skies and the airspace adjacent to land. As such, the air littoral is a bit of a no man's land as most air forces mainly focus on blue skies whereas land-based air defense capabilities to protect this small strip have atrophied.¹⁴ This was a non-issue during the type of military operations constituting the norm in the past two decades, but it is becoming a major issue with the much more frequent use of UAV, First-Person View (FPV) drones, missiles, and artillery all sharing the same airspace.

As the air littoral is becoming more crowded and contested, the US Defense Advanced Research Projects Agency (DARPA) launched the Air Space Total Awareness for Rapid Tactical Execution (ASTARTE) program in 2020. The goal is to "provide a real-time, common operational picture of the dynamic" future airspace.¹⁵

¹⁴ On the contested paradigms informing the division of responsibilities between air and land forces for air littoral operations, see: Bremer/Grieco, "The Air Littoral: Another Look," pp. 75-78.

¹⁵ "Real-time Airspace Awareness and De-confliction of Future Battlespaces."

A common operational picture that helps deconflicting the use of the air littoral is important but not enough.

Today's challenges make it paramount to "reconceptualize air control (as) control of the air littoral is rapidly decoupling from that of the blue skies."¹⁶ This presents an obvious dilemma: losing control of the air littoral cripples air power from below by letting adversaries create a hard-to-penetrate air space while at the same time creating heightened dangers from above for the land forces that encroach upon their freedom of maneuver and drastically reduce their survivability.¹⁷

In addressing dangers in the air littoral, most analyses focus on the advent of UAS swarms. There are heated debates if swarms are going to be successful or not. Justin Bronk and Jack Watling of the Royal United Services Institute (RUSI) correctly argue that developing unmanned platforms for a mass precision strike complex requires demanding tradeoffs about airframe, propulsion and power, navigation, datalinks, sensors, effectors and regulatory aspects such as safety requirements.¹⁸ One question, however, is conspicuously absent from the debate: To what extent will future tactics, developed with the use of AI, shape the battlefield behavior of autonomous swarms and how will these AI tactics shape platform, sensor, effector design and characteristics?

This focus on AI tactics reveals a gap in the literature, as most analysts seem unaware that AI can learn to develop battlefield tactics.¹⁹ That's why the findings of our paper inform military concepts development while at the same time contributing to the broader literature. By discussing how AI can self-learn to develop a tactic like collaborative coordination, we show what fighting in the future air littoral could look like. At the same time, we also demonstrate that the offense vs defense debate needs to be much more nuanced as it is anything but straightforward if attacking swarms or ground-based air defenders will gain the upper hand.

4.2 Wild Hornets Parameters

The vignettes, the specifications of various digital models, and our approach to generating simulation data are important to properly interpret the findings of Wild Hornets. Before going into details, we emphasize that creating a highly detailed simulation environment with a powerful non-cooperative red force remains key to accomplish the study goals of Wild Hornets. If helicopters are expected to launch

16 Bremer/Grieco, "The Air Littoral: Another Look," p. 72.

17 See also: Lucas, "The Drone Dilemma of the U.S. Air Force," Bremer/Grieco, "The Air Littoral: Another Look," p. 80.

18 Bronk/Watling, Mass Precision Strike, pp. 6-21.

19 Borchert, "The Very Long Game of Defense AI Adoption: Introduction," pp. 6-8, 33-34.

ALEs to successfully engage adversarial GBAD, it is paramount for the adversary to challenge the attacking swarm. That is the reason why the building blocks of the simulation have been modeled with outmost care for detail and – even more importantly – for unpredictability to surprise the opponent.

Vignettes

The vignettes and the core research questions have been defined by the ACCDC. Four vignettes describe the overall situation, the specific mission goals as well as different effector combinations.

To start with an Army aviation unit using four attack and three transport helicopters flies around 20km to the front edge of battle area at below <100ft. Its task is to identify and neutralize an adversarial high-value target. For this purpose, helicopters launch several reconnaissance ALEs equipped with optical sensors. The opponent has stationed several Short-Range Air Defense (SHORAD) systems to protect a high value target. Once detected, ALEs attack the SHORAD constellation from different directions to degrade the protection layer that surrounds the high value target. The scenarios modify these core parameters to make the mission increasingly complex, competitive, and thus also more risk-prone.

Digital SHORAD and Munition Models

The SHORAD system has been modeled using the German Gepard Artillery Anti-Aircraft (AAA) system as the reference.²⁰ It can be interpreted as a prototypical gun-based SHORAD system like the Russian 2K22 Tunguska and the 96K6 Pantsir or the Chinese Type08 systems. Still today, the Gepard's overall latency and the powertrains used for the turret and the gun set standards difficult to meet for other gun-based SHORAD systems.²¹

We have also chosen this system because the Gepard's physical parameters (e.g., traverse time of the turret, firing cadence), the internal processing cycle, and the sensor packages are quite well known. We used the KWS FlakPz2, which did not enter service with the German Army. This system has two periscopes plus optical trackers, thermal image, laser rangefinders, an MDPR 12 search radar complemented with a tracking radar and fire control systems.²²

²⁰ See for example: <https://www.knds.de/en/systems-products/tracked-vehicles/air-defence-systems/gepard-1-a2/> (last accessed 28 August 2024).

²¹ For a more general discussion of the role of air defense guns in counter-swarming, see: Flynn, "Counterswarming Is Imperative." We underline that Wild Hornets is focusing on gun-based SHORAD systems. Therefore, it is not possible to transfer our results to missile-based SHORAD systems that were out of scope of the Wild Hornets study.

²² Acoustic sensors for ALE detection have not been considered.

Some scenarios have been calculated with the Gepard using an external Recognized Air Picture (RAP). In addition, it has been assumed that direct Gepard-to-Gepard communication is possible. Rather than using separate search-and-track radars, the digital Gepard is using a combined E-scan radar model provided by GhostPlay team member Hensoldt. This state-of-the-art multifunctional radar operates in search, confirm, and track modes. For our simulation, we have readjusted the radar parameters to approximately match the performance of the original Gepard radar system.

The digital Gepard uses four different types of munitions: High-explosive incendiary (HEI), semi-armor-piercing HEI (SAPHEI), frangible armor piercing discarding sabot (FAPDS), and advanced hit efficiency and destruction (AHEAD) munition. The initial speed of these types of munitions ranges between 1,175 and 1,400m/s with an effective range of 3,500 to 4,500m. For the use of AHEAD munition, the digital model of the Gepard has been equipped with a laser-range finder, which programs the dismantling time of the projectile upon leaving the barrel. To simulate the ballistic curves, we have used a model with quadratic drag.²³

Digital ALE Models

ALE systems have been modeled based on AeroVironment's Switchblade 600 loitering munition (LM).²⁴ We built two physically sufficient simulation models to implement the flight dynamic. One model represents flights with a flight bearing controller; the second model offers non-stabilized flights directly using the actuator control without the flight bearing controller. Although expert pilots considered both models as realistic, we concede that our models are only generic digital representations of the fixed-wing LM. Therefore, we cannot guarantee how accurately the model represents the flight dynamics of a real-world Switchblade 600.²⁵

In addition to the Switchblade 600 we have modeled a Non-Line of Sight (NLOS) missile based on the SPIKE NLOS,²⁶ for which we have built an adequate digital replica. Other types of LM could have been used to construct the ALE swarm, but we consider the Switchblade 600 representative for the research questions under consideration. When technical differences mattered – such as jet propulsion (e.g., Coyote LM²⁷) vs propellers – we have adjusted the specific parameters.

²³ Ballistic models based on the NATO Ballistic Kernel (NABK) are available for use in the Defense Metaverse but have not been used for Wild Hornets as the evaluations were non-classified.

²⁴ For more details, see: <https://www.avinc.com/lms/switchblade-600> (last accessed 28 August 2024).

²⁵ In cooperation with UAV suppliers of the Bundeswehr we have further refined our UAV model to make sure it can be adjusted to the true flight characteristics of different UAVs.

²⁶ For more details, see: <https://www.rafael.co.il/system/spike-nlos/> (last accessed 28 August 2024).

²⁷ For more details, see: <https://www.rtx.com/raytheon/what-we-do/integrated-air-and-missile-defense/coyote> (last accessed 28 August 2024).

The simulation model also includes ALE-NLOS missile communication in the end game. We experimented with different communication models reflecting properties comparable to asynchronous high latency satellite-based communication as well as progressive information propagation via mesh communication using other swarm members as relay nodes.

Digital Sensor Models

Sensor simulation in our Defense Metaverse is based on signal-based, probabilistic, and auto-encoder models. Signal-based models provide a realistic replica of the sensor processing model or parts of it. These models also offer plausible high-resolution sensor images, which can be verified. For our simulation we have used signal-based models provided by GhostPlay partner Hensoldt. But signal-based models require significant computing power to simulate physical modes of action. This creates challenges to compute scenarios that need to run faster than real-time. That's why we have focused on using signal-based models only to analyze specific scenarios.

Sampling and parametrization make it possible to compute probabilistic sensor models that match the performance parameters of signal-based models. Probabilistic sensor models are very well suited for the use in tactical scenario simulations as they need significantly less computing power. They capture major statistical properties of the signal-based models but are not single-case plausible. Probabilistic sensor models have only been used for training runs to develop policies, which were then transferred to operate with signal-based models.

The very high resolution of the GhostPlay Defense Metaverse paves the ground for immersive sensor representation. First, sensor data can be recorded in a real environment to train the auto-encoder. This generates a condensed internal representation of the sensor image commensurate with real conditions. The same location that has been used for recording can also be chosen in the Defense Metaverse to locate the digital sensor at the exact same spot. Then, the simulated environment will replace the original sensor input. The parameters of the auto-encoder will be adapted to replicate the recorded sensor image upon receipt of the input signal. This method also captures the artifacts of the original sensor in the digital environment thus compensating the shortfalls of signal-based sensor models. This, in turn, creates very realistic and highly detailed digital models, which also include realistic ground clutter effects.²⁸

28 For an overview of the different forms of surface, volume or point clutter, see: Wolff, "Radar Clutter."

Simulation Data

A final word is due on simulation data. We have only used open data based on information provided by Janes and our own research. Whenever possible, we have used specific analytical software to generate data relevant for the perception models. For example, the ALE radar cross section (RCS), that was used with the radar models, has been calculated using Matlab Package POfacets.²⁹ We have also used suitable software to simulate and calculate electro-optical (EO) and infrared (IR) images, so that corresponding overlays were available for each object in the respective spectral range. Furthermore, we developed assumptions for the material used to manufacture the ALE systems by assessing publicly available information and analyzing videos of systems crashed and recovered in Ukraine.

Despite great care for detail, we did not use original systems data. However, looking at the simulation findings, we concede that limits accruing from not using original data are negligible. Overall, our simulation findings are well-conditioned, which means that small changes of the parameters lead to small changes in the simulation results. In addition – and more importantly –, the AI system learns how to work with modified parameters and readjusts its behavior commensurate with parameter modifications.

²⁹ Prof. David C. Jenn at the Naval Postgraduate School has developed POfacets as a freely available Matlab script. With the help of the software, monostatic and bistatic RCS can be calculated based on three-dimensional computer-aided design (CAD) objects. POfacets has been specifically designed to engineer radar and laser cross sections. For more, see: <https://faculty.nps.edu/jenn/> (last accessed 28 August 2024).

5 Emergent AI Tactics Make the Difference

5.1 Setting the Baseline

Before looking at the specific AI tactics developed with Wild Hornets, we want to discuss the simulation-based findings of using state of the art ALE against the SHORAD constellation described above.³⁰

Initial tests resembled scenarios that most ALE and SHORAD solution providers currently use to demonstrate the performance of their systems. For these tests, helicopters were launching ALE up to 100ft above ground. In some scenarios helicopters provide ALE guidance, in others, ALE swarm members were performing this function. When attacking the targets, ALE were following a descending trajectory. We have used ALE swarms of different size (up to 50 members) to attack SHORAD systems. SHORAD systems were operating from randomly chosen positions and under different day-light conditions to start with.³¹ This helped us establishing the baseline to identify the success rate in the basic configuration.³²

Based on 10,000 sample scenarios³³ using different SHORAD constellations, our results suggest the following:

- Only in one out of 10 cases will a state-of-the-art ALE using traditional trajectories succeed against the SHORAD constellation shown in the simulation.
- A SHORAD system using active sensors will be even more powerful as these sensors improve the lead time to detect and engage incoming ALE. Tracks can be automatically initialized if the ALE swarm operates out of ground clutter. Electro-optical (EO) sensors compensate occasional track losses.
- The performance of a SHORAD system that is manually operated (human in the loop) and uses passive sensors (and laser) decreases significantly when facing a swarm of three to four ALE approaching from different directions. This is mainly due to the zoom-latency to detect, classify, and identify small flying objects, such as small Unmanned Aerial Systems (sUAS).
- Things change dramatically to the benefit of the defender if the SHORAD system is AI augmented. In this case it is almost impossible for ALE swarms – using conventional trajectories and flying at around 120-180 km/h – to overwhelm the SHORAD constellation. In around 60% of all scenarios, ALE swarms with up to 40 members suffered total destruction.

³⁰ Readers should bear in mind that we generalize our findings and refrain from getting too detailed.

³¹ The Defense Metaverse used for Wild Hornets provides different weather modes such as heavy rain, fog, snow and ice, twilight, and darkness.

³² Success means that at least one ALE was able to penetrate the SHORAD system and explode on the target.

³³ We ran a total of 1.8M scenarios.

Overall, the superior performance of the SHORAD system is due to active and coordinated EO sensor control, active radar tracking, and the Gepard's ultra-short control cycle. This means that with AI control, the total latency of the system can be reduced almost to the system's mechanical latency. In this case the low tempo of the ALE swarm constitutes its main disadvantage.

5.2 AI Tactics for the Endgame

The above results provided the point of departure for further analyses. In line with the requirements of the ACCDC we were interested in identifying the key performance characteristics future ALE swarms need to meet to overwhelm the simulated SHORAD constellations.

Focusing first on the endgame, when the ALE swarm engages the SHORAD, our simulation results demonstrate that terrain following ALE swarms that approach the target using highly agile flight maneuvers stand a significantly higher chance of success than status quo solutions. Most importantly, however, swarms that self-learn how to coordinate, define, and delegate tasks and sub-tasks among individual swarm members create by far the biggest leap ahead that even highly powerful AI-augmented SHORAD systems have a hard time countering.

Terrain Following

For our tests we reconfigured the ALE swarm to become terrain following as shown in Figure 3. This implies that the swarm can adjust the flight altitude to make optimal use of simulated terrain conditions. During the training sessions the control strategy freely set the flight altitude. Neither have we defined specific waypoints nor minimal flight altitudes that the ALE swarm needed to comply with. We only prevented the simulation from overrunning collision with terrain, buildings or vegetation, that caused ALEs to crash.

The existing ground truth provided the basis for the forward-looking coordination function that underpins terrain following. This was an appropriate choice to gauge the required minimal flight altitude that the ACCDC wanted to identify. As the simulator included detailed models of vegetation and infrastructure (e.g., powerlines), standard algorithms could be used to implement dynamic collision detection between objects and environment. We have used Bayesian models to simulate timely recognition of obstacles and turbulences in low altitude flights and obtain realistic loss rates based on flying altitudes. For example, the probability of collision with an object depends on the probability that the object has inadvertently not been identified. This, in turn, depends on aspects like distance, tempo

Figure 3: Terrain Following ALE Swarm Approaching Targets



of the overflight and sensor vision. Moreover, effects like doppler notches were implemented, which allowed ALEs to provoke track drops, which in turn increased the SHORAD latency to engage the target. Thus, the resulting control policies implement optimal trade-offs between low altitude approaches and the risk of early detection.

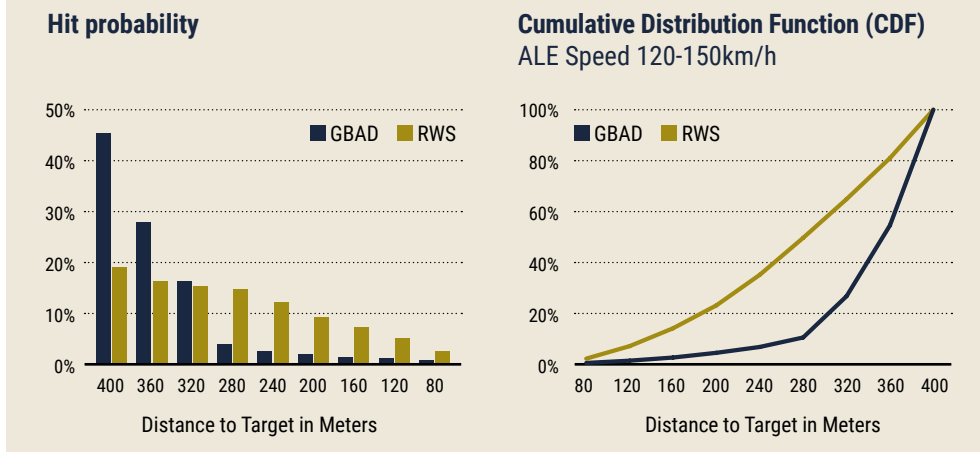
Our overall finding is that terrain following formations challenge SHORAD systems as this flight pattern is more difficult to detect. This can – in principle – offset the fact that small caliber munition can engage low and slow flying ALE. But if ALE swarms can exploit terrain features, they are able to lower the flight altitude thereby reducing the time available for the SHORAD system to detect the swarm.

The outcome of this type of maneuver is significant, as Figure 4 illustrates. Here we compare the hit probability of ALE swarms against a Gepard-based SHORAD system (blue) and a Remote Weapon Station (RWS, yellow):³⁴

- First, our results suggest that in 7 out of 10 cases ALE swarms will succeed against the RWS, if the swarm succeeds in coming as close as 180m to the target without being detected. By contrast, the high rotation rate of the Gepard's turret and weapon further cut the distance to the incoming target to 110m.

³⁴ RWS have been modelled based on currently available systems.

Figure 4: ALE Swarm Mission Success against Gepard-based SHORAD Systems (blue line) and Remote Weapon Stations (yellow line)



- Second, these results suggest that current scenarios used to demonstrate the performance of C-sUAS solutions are not realistic.³⁵ In many cases, these scenarios assume that stationary flying objects (like quadcopters) can be successfully countered at several hundred if not thousand meters. However, this scenario will hardly materialize because the standoff constellation is too detrimental for the sUAS to succeed. Consequently, C-sUAS demonstrations need to include scenarios in which sUAS approach the countering systems at up to 180m or below.

To sum up, ALE swarms that learn how to exploit terrain features, vegetation, building properties, and weather effects stand a high chance of knocking out many types of gun-based SHORAD systems currently in use. Moreover, ALE swarms that know how to confuse the SHORAD sensor suite, for example by hiding in ground clutter, further improve their rate of survival.

What can be done to improve the SHORAD's survival rate? Several options are available. In our scenarios, we have used a fully automated Gepard with AI control, which scales nicely against different swarm sizes. Again, the low overall latency time improves the success rate of the AI-Gepard even against low and lowest flying ALE swarms. Further augmenting the SHORAD with our proprietary track-by-attention method makes it even more robust. This method controls the

³⁵ See for example the following promotional videos available on YouTube: https://www.youtube.com/watch?v=pb5_F4_Eod8; <https://www.youtube.com/watch?v=y4d2d77Rtfl&t=1s> (last accessed: 27 August 2024).

effector by directly using input data and, instead of pursuing individual targets, identifies target clusters that can be engaged with barrage fire thereby exploiting the recoil of the Gepard's twin cannon.³⁶ Track-by-attention thus provides a potential solution out of the 'Muhammed Ali dilemma' discussed in chapter 2.

Collaborative Self-Coordination

Using different attack vectors to engage the SHORAD constellation with numerous ALEs requires coordination.³⁷ Across all scenarios, collaborative self-coordination proved to be the most successful feature for ALE swarms to knock out SHORAD systems. This includes the ability to form smaller sub-teams that will perform individual and collective tasks to accomplish the mission. To do so, the ALE swarm must develop ad hoc strategies that define mission goals and sub goals and delegate the execution of these goals to team members. Algorithmically speaking, this can be done with distributed control, and we have used decentralized partially observable Markov Decision Processes (decPOMDP) to model collaborative coordination.³⁸

It is very important to understand that collaborative self-coordination, as depicted in Figure 5 with a swarm of three UAS, has nothing to do with traditional swarming and goes way beyond the patterns currently observed in Ukraine.³⁹ Existing concepts build on the idea that swarm members mimic each other's behavior to fly specific patterns or execute a pre-planned flight direction. By contrast, distributed control implies that each swarm member follows its own goal. If and to what extent swarm members cooperate depends on the probability that cooperation will make it easier to achieve the mission goal. This requires the swarm to self-learn the best tactic to accomplish the relevant mission – during the end game and in the 20km-runup from the helicopter launch to hitting the target. Collaborative self-coordination yields superior results in the simulation:

- For a manually operated Gepard, swarms with up to only 6 entities using collaborative time-sensitive coordination are almost impossible to counter. In 8 out of 10 cases a guided ALE swarm able to exploit terrain features needs 7 members to outpace the Gepard. But with collaborative coordination, the size of the ALE swarm shrinks to four members.

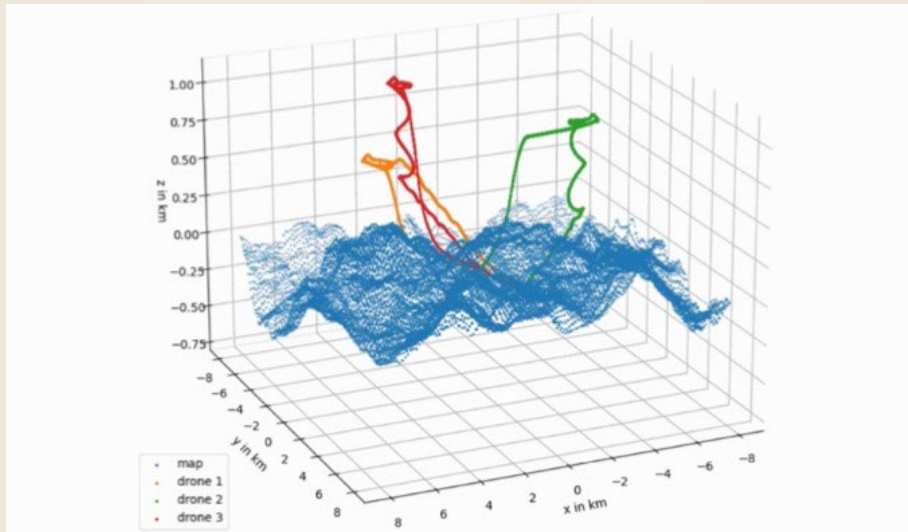
36 Borchert/Brandhuber/Brandstetter/Schaal, *Free Jazz on the Battlefield*, p. 14; Brandhuber, "Superiority on the Battlefield with Tactical AI and Emergence," p. 9.

37 If sequential or simultaneous attacks are more successful very much depends on adversarial SHORAD constellations and air defense tactics. Sequencing is potentially unfavorable if adversarial air defenders dynamically distribute air defense tasks among themselves. By contrast sectoral areas of responsibility increase the ALE swarm's probability of success as targeting can focus on one system after the other. Overall simultaneous attacks from different directions are more promising, simulation results suggest.

38 The decPOMDP was solved by a reinforcement learning method.

39 "In the Ukraine War, drones have operated in stacks rather than swarms. (...) The drone stacks in the war in Ukraine have been coordinated through multiple drone operators using software-based battle networks or traditional means of communication." Pettyjohn, *Evolution Not Revolution*, pp. 39-40.

Figure 5: ALE Swarm with Collaborative Self-Coordination During a Reconnaissance-Strike Mission



- The performance is even more impressive if the swarm attacks a modern and automated SHORAD system. Against this type of defender, a guided ALE swarm flying low and slow was successful only in four out of 10 missions. But ALE swarm members that use collaborative coordination will achieve mission success in 8 out of 10 cases.

These results are significant, because collaborative self-coordination reduces the swarm size – thus lowering the number of swarm members needed and reducing the logistical footprint – while simultaneously increasing the swarm’s combat effectiveness. This illustrates how an AI tactics kernel can provide an edge for a future unmanned ecosystem (see chapter 6).

Again, the question is: Can the SHORAD respond? Our tentative response is yes, but for this we had to fundamentally reconceptualize and redesign the SHORAD’s target recognition and fire control concept. A new approach using game-theoretic paradigms can identify complex swarm tactics at the outset, anticipate future moves, and respond commensurate with changing aggressor tactics. Simulation results suggest that an ALE swarm with around 12 members is needed to succeed in four out of 10 cases against an AI-augmented Gepard using the track-by-attention method. But if the Gepard further uses the new game-theoretic AI tactic, the success rate of the attacker is cut by half to only two out of 10 cases.

Figure 6: Illustrative Patterns of Agile Flight Maneuver in the End Game



Agile Flight Maneuvers

Collaborative coordination enables swarms to fly combat jumps or S-like crossings (Figure 6). The main purpose of these flight patterns is to make it more difficult for the SHORAD to track and trace the ALE swarm. In doing so, swarms can attempt to hide in ground clutter, abruptly change the flight path/direction, or move into an attack position that renders the SHORAD weapon system ineffective (beyond elevations of 80°) or exploits weaknesses of the SHORAD's armor.

For the scenarios we have trained agile flight maneuvers in combination with other patterns. Moreover, we have been cooperating with professional drone racers to record their flight patterns for the end game with the goal to stretch the aerodynamic features of commercial-off-the-shelf FPV drones to the maximum. We used these trajectories to improve the selectivity of the low-level controller, reduce training efforts, and integrate extreme flight patterns into our AI tactics, that are hard to explore because they are more brittle than traditional flight patterns. Initial explorations with these patterns suggest that the ALE swarm's success rate can be further advanced, in particular when facing a fully automated and AI-augmented SHORAD system.

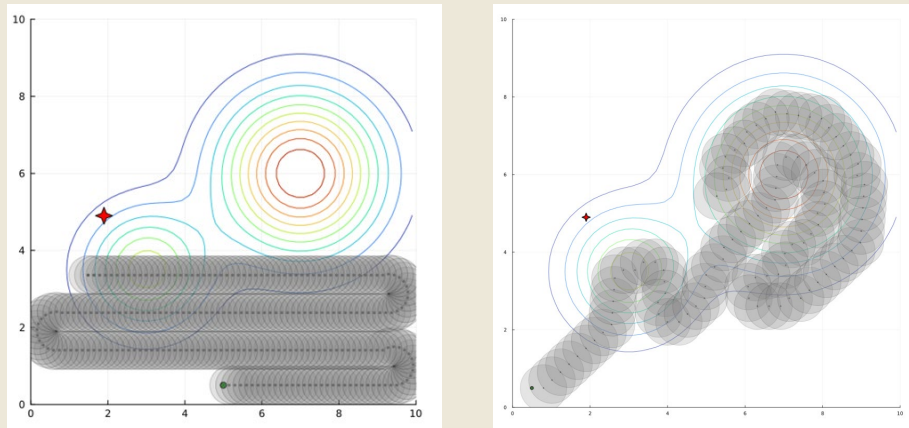
5.3 AI Tactics for Reconnaissance-Strike Missions

In most cases enemy positions are unknown or hard to detect because of camouflage, signature management, and other measures undertaken to hide. That's why reconnaissance missions need to survey the relevant mission area and identify and confirm enemy positions before the ALE swarm enters the end game.

Ideally, the ALE swarm learns to develop a reconnaissance tactic commensurate with the threat landscape and the mission goals. Most importantly, this tactic should be unpredictable and hard (if not impossible) for the adversary to recognize and detect. Only if the aggressor ALE swarm remains unnoticed for as long as possible will it be able to approach the SHORAD inside the 'dead zone' as discussed above.

Currently, most reconnaissance missions emulate the so-called sweep pattern, which has been developed for search and rescue (SAR) operations.⁴⁰ As Figure 7 (left) illustrates, the reconnaissance object flies in a regular pattern over the target – thereby moving like a lawn mower – until the SHORAD location (illustrated by the red star) has been identified. This pattern is effective and efficient – as long as there are no non-cooperative objects on the ground.

Figure 7: Sweep (left) vs MDP-based (right) Reconnaissance Patterns



⁴⁰ See for example: Kratzke/Stone/Frost, "Search and Rescue Optimal Planning System."

On the battlefield, however, adversarial non-cooperation is the norm. That's why the sweep pattern is inadequate for combat reconnaissance. It is too easily detectable and reveals an aggressor's intentions. Once the SHORAD has detected the pattern, it is easy to anticipate future moves, which improves kill rates. In doing so, defenders do not even have to switch on active sensors, which prevents them from being detected. Therefore, we have developed a new reconnaissance tactic using Markov Decision Processes (MDP) to produce a pattern that appears random and unpredictable (Figure 7, right).

What looks weird at first sight turns out to be powerful when assessing the findings of the statistical analysis. If you compare average reconnaissance time with the number of targets to be detected and vary assumptions for false positives and non-detection, it becomes clear that this pattern fully leverages the 'Muhammed Ali dilemma' to the benefit of the attacker, because it is much more challenging to understand. In addition, our new MDP reconnaissance pattern also blends prior knowledge about the target area with information about the current situation on the ground. In the end, the MDP-based tactic not only detects SHORAD positions more reliably but also much faster than the sweep approach.

Combining collaborative coordination with MDP-based reconnaissance tactics turns the reconnaissance swarm into a reconnaissance-strike swarm with some ALE swarm members temporarily acting in reconnaissance mode, while the strike members are already loitering above the SHORAD at low altitude. This combination yields an autonomous and interactive reconnaissance-strike swarm tactic.⁴¹ This tactic is superior to existing approaches because it

- effectively uses prior information, for example, from electronic support measures (ESM), Synthetic Aperture Radars (SAR),⁴² intelligence on adversarial SHORAD systems as well as terrain specific aspects such as vegetation, building properties, trafficability, and tactical terrain evaluation;⁴³
- advances combat efficiency and effectiveness as it considers different search criteria;⁴⁴
- allows for the use of heterogeneous swarms with fixed wing and rotary UAS, as both types of assets use different flight trajectories and thus need different tactics that mirror these specific aspects;

41 By "autonomous" we refer to a mission with different levels of human engagement. Reconnaissance and strike elements directly interact with each other without relying on traditional C2 loops and reach-back to headquarters. Human operators, however, remain involved because they define the overall mission goals and the impact to be achieved. Within these parameters the developed reconnaissance-strike tactic allows the ALE swarm to act independently.

42 We are currently working on integrating SAR-based information for target assignment.

43 Tactical terrain evaluation provides information about the presumed locations of SHORAD system the adversary would likely use to achieve the best kill rates against the ALE swarm.

44 If there are lakes or rough terrain in the target area under consideration these areas can be surpassed quite quickly as it is relatively unlikely that an adversary would use these spots as SHORAD locations.

- considers the fact that adversaries can relocate SHORAD systems by developing hypotheses⁴⁵ regarding possible future locations to readjust the reconnaissance tactic.

5.4 AI Tactics for Target Assignment

Autonomous target assignment for the reconnaissance-strike swarm is a specific element that can increase mission success. We have looked at the idea of using ALEs to guide NLOS missiles to hit adversarial SHORAD systems. Two basic patterns emerge: First, loitering ALEs that perform reconnaissance or strike missions learn to guide the NLOS missile by sharing their target coordinates and flight trajectories with them, as illustrated in Figure 8. Second, ALEs can operate as 'door openers.' In this function they hit adversarial sensor posts first. This creates gaps in the adversarial air defense posture that NLOS missiles can use to sneak in and hit the ultimate targets.

Figure 8: ALE Used for NLOS Missile Target Assignment



⁴⁵ Tactical terrain evaluation, prior locations used to successfully engage ALE swarms, intelligence about adversarial SHORAD systems and other considerations play a role in developing these hypotheses.

Based only on the Wild Hornets scenarios it is not possible to reliably gauge the number of ALEs needed for mission success or decide if sequential or simultaneous attacks are more successful.⁴⁶ However, we have witnessed a specific pattern that is most interesting given the opportunistic use of assets, which adds to the unpredictability of the heterogenous ALE-missile swarm: If the mission-specific attrition rate of the ALE swarm members is too high, i.e., the swarm loses too many ALEs to be successful, ALEs learn to call in NLOS missiles as temporary backups that are more powerful against the target. This tactic could provide an option to engage low altitude acquisition radar systems like the Russian 76N6 Clam Shell.⁴⁷ Simulations reveal that attacking the 76N6 Clam Shell from below is not feasible, because the kinetic energy of the low-flying ALEs is insufficient to ignite the charge when attacking. Furthermore, this maneuver will make ALEs an easy target for the SHORAD systems that protect the 76N6 Clam Shell. However, the ALE-NLOS missile combination could offer a solution.

⁴⁶ The main reason is that ALE swarms did the job without the need for missiles.

⁴⁷ For more on this system, see: <https://www.ausairpower.net/clamshell.html> (last accessed 28 August 2024):

6 Outlook: An AI Tactics Kernel for Software- Defined Defense

The advanced concepts study Wild Hornets, conducted in cooperation with the German ACCDC, illustrates the benefits of using the Defense Metaverse for force development. Our findings show: AI tactics that exploit terrain conditions (terrain following) and advance collaborative coordination can make future ALE swarms much more powerful. In response, AI tactics leveraging insights from game theory can empower air defenders to fend off these swarms.

Right now, the AI tactics developed in the Defense Metaverse are gradually transferred into the physical environment. Together with the Bundeswehr Research Institute for Materials, Fuels, and Lubricants (Wehrwissenschaftliches Institut für Werk- und Betriebsstoffe), we are working on generating adapted tactics that will be used to equip the institute's future ecosystem of unmanned systems. If successful in the real-life setting, these AI tactics will be deployed for operations.

This approach is in line with the German Bundeswehr's emphasis on software-defined defense to augment legacy systems and advance armed forces transformation with the help of synchronized hardware and software development.⁴⁸ AI tactics, as argued in this paper, should be interpreted as the core of future software-defined defense solutions. Mature armed forces that already use different mission planning/support systems to integrate their assets into the sensor-to-shooter cycle can use the AI tactics kernel as an overlay system that fully respects and interacts with legacy software. Armed forces operating at the lower end of the maturity spectrum can use the AI tactics kernel to leapfrog by letting the kernel take over the function of traditional mission planning systems.

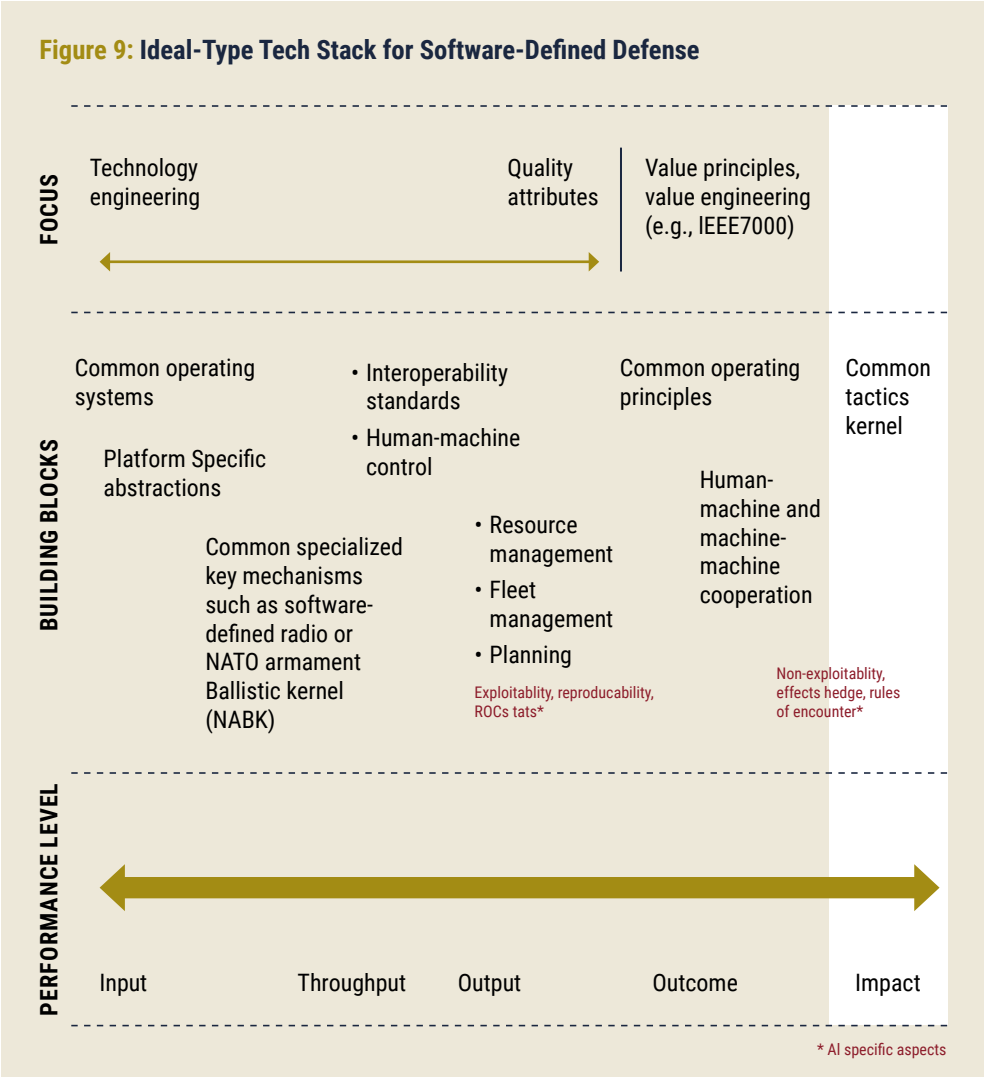
As illustrated in Figure 9, the portfolio of software-defined defense solutions is broad. However, most of today's solutions are input-driven because they emphasize data as a key resource armed forces need to deal with. In contrast, the concept of an AI tactics kernel is impact driven: Rather than asking what is available to accomplish a military mission, the AI tactics kernel asks what impact the mission is expected to deliver. The two perspectives are complementary but emphasize different aspects.

This distinction also paves the ground for a new tactics stack to complement the technology stack of software-defined defense. The tactics stack offers novel ways to finetune and deliver military power by interpreting tactics as the value layer that armed forces, which use a service-oriented architecture, can move across domains and operating levels. The tactics stack advances battlefield versatility by coordinating the behavior of troop elements, platforms, and payloads like sensors and effectors. In addition, the tactics stack will be most instrumental in coordinat-

⁴⁸ Borchert/Schütz/Verbovszky, "Master and Servant: Defense AI in Germany," p. 201.

ing interactions of platform-payload ecosystems, first, within and across domains and, second, with human operators. To this purpose, the AI tactics kernel needs to be interoperable with legacy and future software applications. The Defense Metaverse will help train the respective interactions and highlight potential flaws in the respective workflows. Lessons identified from ongoing operations and intelligence information regarding adversarial forces can be integrated via the training environment of the Defense Metaverse thus paving the ground for continuous software-defined development in line with current national and allied capability demands, ongoing battlefield requirements, and visions of future forces.

Figure 9: Ideal-Type Tech Stack for Software-Defined Defense



Literature

"Real-time Airspace Awareness and De-confliction for Future Battles," DARPA press release, 4 July 2020, <https://www.darpa.mil/news-events/2020-04-07> (last accessed 1 July 2024).

Arthur, Gordon, "US approves loitering munitions sale for Taiwan's 'porcupine strategy,'" C4ISRNet, 20 June 2024, <https://www.c4isrnet.com/battle-field-tech/2024/06/20/us-approves-loitering-munitions-sale-for-taiwans-porcupine-strategy/> (last accessed 28 August 2024).

Borchert, Heiko, "The Very Long Game of Defense AI Adoption: Introduction," in Heiko Borchert, Torben Schütz, and Joseph Verbovsky (eds.) *The Very Long Game. 25 Case Studies on the Global State of Defense AI* (Cham: Springer Nature, 2024), pp. 1-38, https://doi.org/10.1007/978-3-031-58649-1_1 (last accessed 28 August 2024).

Borchert, Heiko, Torben Schütz, and Joseph Verbovsky, "Mater and Servant: Defense AI in Germany," in Heiko Borchert, Torben Schütz, and Joseph Verbovsky (eds.) *The Very Long Game. 25 Case Studies on the Global State of Defense AI* (Cham: Springer Nature, 2024), pp. 197-218, https://doi.org/10.1007/978-3-031-58649-1_9 (last accessed 28 August 2024).

Borchert, Heiko, Christian Brandhuber, Armin Brandstetter, and Gary S. Schaal, *Free Jazz on the Battlefield. How GhostPlay's AI Approach Enhances Air Defense*. DAIO Study 22/03 (Hamburg: Defense AI Observatory, 2022), https://defenseai.eu/daio_study2203 (last accessed 28 August 2024).

Brandhuber, Christian, "Superiority on the Battlefield with Tactical AI and Emergence. The Next Generation," NATO Science and Technology Organization, Lecture Series on AI, SET-290-RLS.

Bremer, Maximilian K. and Kelly A. Grieco, "The Air Littoral: Another Look," *Parameters*, 51:4 (Winter 2021-22), pp. 67-80, <https://press.armywarcollege.edu/parameters/vol51/iss4/7/> (last accessed 28 August 2024).

Bronk, Justin and Jack Watling, *Mass Precision Strike. Designing UAV Complexes for Land Forces* (Lon-

don: RUSI, 2024), <https://www.rusi.org/explore-our-research/publications/occasional-papers/mass-precision-strike-designing-uav-complexes-land-forces> (last accessed 28 August 2024).

Brose, Christian, *The Kill Chain. Defending America in the Future of High-Tech Warfare* (New York: Hachette Books, 2020).

Clark, Bryan, Dan Patt, and Timothy A. Walton, *Implementing Decision-Centric Warfare: Elevating Command and Control to Gain an Optionality Advantage* (Washington, DC: Hudson Institute, 2021), <https://www.hudson.org/national-security-defense/implementing-decision-centric-warfare-elevating-command-and-control-to-gain-an-optionality-advantage> (last accessed 28 August 2024).

Clark, Bryan, Dan Patt, Harrison Schramm, *Mosaic Warfare. Exploiting Artificial Intelligence and Autonomous Systems to Implement Decision-Centric Operations* (Washington, DC: CSBA, 2020), <https://csbaonline.org/research/publications/mosaic-warfare-exploiting-artificial-intelligence-and-autonomous-systems-to-implement-decision-centric-operations> (last accessed 28 August 2024).

Digital Engineering Strategy (Washington, DC: Office of the Deputy Assistant Secretary of Defense for Systems Engineering/Department of Defense, 2018), <https://apps.dtic.mil/sti/citations/AD1068564> (last accessed 28 August 2024).

Flynn, Karl, "Counterswarming Is Imperative," *Proceedings*, 150:8 (August 2024), <https://www.usni.org/magazines/proceedings/2024/august/counterswarming-imperative> (last accessed 28 August 2024).

Freedman, Lawrence. *Strategy. A History* (Oxford: Oxford University Press, 2013).

Grayson, Tim, "Mosaic Warfare," DARPA presentation, 27 July 2018, <https://www.almendron.com/tribuna/wp-content/uploads/2019/07/sto-mosaic-distro-a.pdf> (last accessed 28 August 2024).

Haystead, John, "DARPA's Mosaic Warfare: Moving to Address the Ever-More-Rapidly Paced Advances/ Changes in Fielded Threat Capabilities," *The Journal of Electronic Defense* (February 2020), pp. 20-24.

Hicks, Kathleen, "'Unpacking the Replicator Initiative' at the Defense News Conference," Department of Defense, 6 September 2023, <https://www.defense.gov/News/Speeches/Speech/Article/3517213/deputy-secretary-of-defense-kathleen-hicks-remarks-unpacking-the-replicator-initi/> (last accessed 28 August 2024).

Johnston, Chris, "Muhammad Ali's best quotes: 'Float like a butterfly, sting like a bee.'" *The Guardian*, 4 June 2016, <https://www.theguardian.com/sport/2016/jun/04/muhammad-ali-greatest-quotes-sting-butterfly-louisville-lip> (last accessed 28 August 2024).

Kratzke, Thomas, Lawrence D. Stone, and John R. Frost, "Search and Rescue Optimal Planning System," 13th International Conference on Information Fusion, Edinburgh, 26-29 July 2010, <https://doi.org/10.1109/ICIF.2010.5712114> (last accessed 28 August 2024).

Lucas, Clifford, "The Drone Dilemma of the U.S. Air Force," *War on the Rocks*, 7 May 2024, <https://warontherocks.com/2024/05/the-drone-dilemma-and-the-u-s-air-force/> (last accessed 28 August 2024).

Magid, Jacob and Amy Spiro, "April 13: Iran fires 300 missiles and drones; most intercepted; minor damage at IDF base," *Times of Israel*, 13 April 2024, <https://www.timesofisrael.com/liveblog-april-13-2024/> (last accessed 28 August 2024).

Pettyjohn, Stacie, *Evolution Not Revolution. Drone Warfare in Russia's 2022 Invasion of Ukraine* (Washington, DC: CNAS, 2024), <https://www.cnas.org/publications/reports/evolution-not-revolution> (last accessed 28 August 2024).

Stephenson, Neal, *Snow Crash* (New York: Bantam Books, 1992).

Targeting. United States Air Force Doctrine Publication 3-60 (Maxwell AFB: United States Air Force, 2021).

https://jdeis.js.mil/jdeis/new_pubs/jp3_60.pdf (last accessed 28 August 2024).

Wolff, Christian, "Radar Clutter," undated, <https://www.radartutorial.eu/11.coherent/co04.en.html> (last accessed 28 August 2024).

Work, Robert O. and Shawn Brimley, 20YY: *Preparing for War in the Robotic Age* (Washington, DC: CNAS, 2014), <https://www.cnas.org/publications/reports/20yy-preparing-for-war-in-the-robotic-age> (last accessed 28 August 2024).

Defense AI Observatory Studies

- 24|25** Mahmoud Javadi, Heavy Thunder, No Rain: Defense AI in Iran
- 24|24** Kitsch Liao, Intelligent National Defense Amid Strategic Ambiguity? Defense AI in Taiwan
- 24|23** Motohiro Tsuchiya, Overcoming the Long Shadow of the Past. Defense AI in Japan
- 24|22** Vitaliy Goncharuk, Survival of the Smartest? Defense AI in Ukraine
- 24|21** Youngwook Park, Will the One Ring Hold? Defense AI in South Korea
- 23|20** Tomas Jermalivičius, Caught Between Today and Tomorrow. Defence AI in Estonia
- 23|19** Nikolaos Karampekios, Konstantinos Sakalis, and Iraklis Oikonomou, Harnessing the Potential. Defense AI in Greece
- 23|18** Andreas Graae, Servers Before Tanks? Defence AI in Denmark
- 23|17** Kévin Martin and Lucie Liversain, A Winding Road Before Scaling-Up? Defense AI in France
- 23|16** Sami O. Järvinen, Cautious Data-Driven Evolution. Defence AI in Finland
- 23|15** Inbar Dolinko and Liran Antebi, Embracing the Organized Mess. Defense AI in Israel
- 23|14** Alastair Finlan, A Fertile Soil for AI? Defense AI in Sweden
- 23|13** John Lee, "Overtaking on the Curve?" Defense AI in China
- 23|12** Heiko Borchert, Torben Schütz, and Joseph Verbovsky, Master and Servant. Defense AI in Germany
- 23|11** Katarzyna Zysk, High Hopes Amid Hard Realities. Defense AI in Russia
- 23|10** Yvonne Hofstetter and Joseph Verbovsky, How AI Learns the Bundeswehr's "Innere Führung." Value-Based Engineering with IEEE7000™-2021
- 23|09** Robert C Engen, When the Teeth Eat the Tail: A Review of Canada's Defence Artificial Intelligence
- 23|08** Çağlar Kurç, Enabling Technology of Future Warfare. Defense AI in Turkey
- 23|07** Lauren A. Kahn, Risky Incrementalism. Defense AI in the United States
- 22|06** Yvonne Hofstetter, Wie KI Innere Führung lernt. Wertbasierte Technik mit IEEE7000™-2021
- 22|05** Andrea Gilli, Mauro Gilli, and Ivan Zaccagnini, Exploring the Benefits of a New Force Enabler: Defense AI in Italy
- 22|04** Kenneth Payne, Bright Prospects – Big Challenges. Defense AI in the United Kingdom
- 22|03** Heiko Borchert, Christian Brandlhuber, Armin Brandstetter, and Gary S. Schaal, Free Jazz on the Battlefield. How GhostPlay's AI Approach Enhances Air Defense
- 22|02** Peter Layton, Evolution not Revolution. Australia's Defence AI Pathway
- 21|01** Heiko Borchert, Torben Schütz, Joseph Verbovsky, Beware the Hype. What Military Conflicts in Ukraine, Syria, Libya, and Nagorno-Karabakh (Don't) Tell Us About the Future of War

